Shesh Budhabhatti

budhabhatti.s@northeastern.edu | (860) 716-1711 | CT/MA | www.linkedin.com/in/sheshbudhabhatti | github.com/sheshbud | sheshbud.github.io/portfolio

EDUCATION

NORTHEASTERN UNIVERSITY (Dean's List at Khoury College of Computer Sciences) — **Boston, MA**
Bachelor of Science, Combined Major in Cybersecurity and Business Administration (Finance Concentration) — April 2028
Coursework: Object-Oriented Design, Algorithms & Data Structures, CS Systems, Security Systems, Discrete Structures

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, SQL, Assembly
Systems & Tools: Git/GitHub, Linux/Unix, Docker, GDB, Valgrind, Azure DevSecOps, Jira, Bitbucket, Polarion, Miro, Arduino
Security: Wireshark, Metasploit, Hashcat, threat modeling, penetration testing, cryptographic protocols
Spoken: English, Gujarati, Hindi, Spanish

WORK EXPERIENCE

MEDTRONIC — **Boston, MA**
Software Architecture Co-op — August 2026 – Present
- Building an automated pipeline that auto-generates UML diagrams and design docs from code updates for Medtronic's Hugo™ surgical robot platform
- Unifying design docs, diagrams, and database records into one pipeline, migrating documentation to a core database with a cross-team API

Product Security Intern — June 2026 – August 2026
- Designed and built an autonomous Wi-Fi resilience testbed (Raspberry Pi 5 access point, ESP32 client/interference nodes) and a from-scratch decision engine fusing latency, jitter, packet loss, and signal strength into a live health score
- Validated autonomous failover under load (28 clients, 600 packets/sec) and induced interference (0 to 4 disruptors: latency 3ms to 33ms, 0% packet loss), holding a stable band switch unattended for ~10 hours

MICROSOFT — **Boston, MA**
AI Student Ambassador & WICxDevSecOps Program Participant — October 2025 – February 2026
- Selected for security training on Azure DevSecOps pipelines, threat modeling, and AI model security protocols
- Piloted Copilot/Omi AI with 100+ users, collecting feedback on quality, privacy risks, and edge cases for product teams

MEDTRONIC — **North Haven, CT**
R&D Software Engineering Intern — June 2025 – August 2025 | July 2024 – August 2024
- Reduced test fixture calibration time by 37% through Python automation scripts, saving 4+ hours weekly
- Authored fixture manual for cross-functional teams across US/India (15+ engineers), reducing onboarding time
- Implemented AES-256 and Blowfish encryption protocols for Signia™ Stapling System, ensuring HIPAA compliance and passing FDA cybersecurity audit requirements

SECURE & ASSURED INTELLIGENT LEARNING LAB — **New Haven, CT**
Cybersecurity & Safety Researcher — June 2023 – February 2024
- Identified 8 security vulnerabilities in LLM systems through 50+ prompt injection attacks
- Documented attack patterns and mitigation strategies, presenting findings to lab researchers

KPMG — **Hartford, CT**
Summer Intern, Audit/Tax/Advisory Accounting — July 2023 – August 2023
- Collaborated with 5-person teams across 3 companies, identifying 2 key control weaknesses in inventory tracking
- Analyzed financial data across 3 client studies with Tableau, visualizing audit risk insight for senior stakeholders

G.U.R.L.S CHESS CLUB NON-PROFIT ORGANIZATION — **USA**
Founder & Executive Director & TEDx Speaker — December 2021 – Present
- Founded chess non-profit coaching 200+ students (70% female); delivered 70+ lessons, TEDx talk/USCF feature
- Redesigned curriculum using retention data analysis, improving participant retention from 40% to 78%

PROJECTS

FUSE Filesystem Implementation | C, Systems Programming — **November 2025**
- Built a custom filesystem supporting large files (>4KB) with inode management and mmap/munmap allocation
- Debugged memory corruption using GDB and Valgrind, achieving 31/31 test cases.

Memory Allocator & Data Structures | C, Systems Programming — **October 2025**
- Implemented malloc/free using mmap/munmap with free list management and a first-fit allocation strategy
- Optimized memory reuse through block coalescing, merging adjacent free blocks to minimize fragmentation

BGP Router Implementation | Python, Networking — **February 2026**
- Implemented a BGP router handling handshake, update, and withdrawal messages with longest-prefix-match forwarding and a full tie-breaking chain
- Wrote all IP address bit-math from scratch with no external libraries; rebuilt the forwarding table on withdrawal for correctness

AWARDS/PUBLICATIONS/CERTIFICATIONS

TEDx: "Empowering Equality, Inspiring Action" (2023) | SMARTA Podcast: "If You're Young, You Still Can Make a Difference" (2025) | Lean Six Sigma Yellow Belt (2024) | CT Women's Chess Champ. (2018) | TRAIN's Intro. to Artificial Intelligence Certified (2024)